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Security Assessment Report

XOX Token

24 Feb 2026

This security assessment report was prepared by SolidityScan.com, a cloud-based Smart Contract Scanner.

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01. Vulnerability Classification and Severity

Description

To enhance navigability, the document is organized in descending order of severity for easy reference. Issues are categorized as *Fixed*, *Pending Fix*, or *Won't Fix*, indicating their current status. *Won't Fix* denotes that the team is aware of the issue but has chosen not to resolve it. Issues labeled as *Pending Fix* state that the bug is yet to be resolved. Additionally, each issue's severity is assessed based on the risk of exploitation or the potential for other unexpected or unsafe behavior.

Critical

The issue affects the contract in such a way that funds may be lost, allocated incorrectly, or otherwise result in a significant loss.

High

High-severity vulnerabilities pose a significant risk to both the Smart Contract and the organization. They can lead to user fund losses, may have conditional requirements, and are challenging to exploit.

Medium

The issue affects the ability of the contract to operate in a way that doesn't significantly hinder its behavior.

Low

The issue has minimal impact on the contract's ability to operate.

Informational

The issue does not affect the contract's operational capability but is considered good practice to address.

Gas

This category deals with optimizing code and refactoring to conserve gas.

02. Executive Summary

XOX Token

Uploaded Solidity File(s)

Language	Audit Methodology	Website
Solidity	**Static Scanning**	-
Publishers/Owner Name	Organization	Contact Email
-	-	-

Security Score is GREAT

The SolidityScan score is calculated based on lines of code and weights assigned to each issue depending on the severity and confidence. To improve your score, view the detailed result and leverage the remediation solutions provided.

This report has been prepared for XOX Token using SolidityScan to scan and discover vulnerabilities and safe coding practices in their smart contract including the libraries used by the contract that are not officially recognized. The SolidityScan tool runs a comprehensive static analysis on the Solidity code and finds vulnerabilities ranging from minor gas optimizations to major vulnerabilities leading to the loss of funds. The coverage scope pays attention to all the informational and critical vulnerabilities with over 700+ modules. The scanning and auditing process covers the following areas:

Various common and uncommon attack vectors will be investigated to ensure that the smart contracts are secure from malicious actors. The scanner modules find and flag issues related to Gas optimizations that help in reducing the overall Gas cost It scans and evaluates the codebase against industry best practices and standards to ensure compliance It makes sure that the officially recognized libraries used in the code are secure and up to date.

The SolidityScan Team recommends running regular audit scans to identify any vulnerabilities that are introduced after XOX Token introduces new features or refactors the code.

03. Findings Summary

XOX Token
File Scan

Security Score	Scan duration	Lines of code
91.70/100	**23 secs**	**324**

0	0	1	12	64	24
Crit	High	Med	Low	Info	Gas

This audit report has not been verified by the SolidityScan team. To learn more about our published reports. click here

ACTION TAKEN

0	0	0	103
Fixed	*False Positive*	*Won't Fix*	*Pending Fix*

S. No.	Severity	Bug Type	Instances	Detection Method	Status
M001	Medium	UNCHECKED ARRAY LENGTH	4	Automated	*Pending Fix*
L001	Low	BALANCE EQUALITY	1	Automated	*Pending Fix*
L002	Low	LACK OF ZERO VALUE CHECK IN TOKEN TRANSFERS	1	Automated	*Pending Fix*
L003	Low	MISSING EVENTS	5	Automated	*Pending Fix*
L004	Low	MISSING ZERO ADDRESS VALIDATION	2	Automated	*Pending Fix*
L005	Low	NONREENTRANT MODIFIER PLACEMENT	2	Automated	*Pending Fix*
L006	Low	OUTDATED COMPILER VERSION	1	Automated	*Pending Fix*
I001	Informational	ADDING A RETURN STATEMENT WHEN THE FUNCTION DEFINES A NAMED RETURN VARIABLE IS REDUNDANT	1	Automated	*Pending Fix*
I002	Informational	BLOCK VALUES AS A PROXY FOR TIME	4	Automated	*Pending Fix*
I003	Informational	CONSIDER USING UINT48 FOR TIME-RELATED VARIABLES	1	Automated	*Pending Fix*
I004	Informational	CONTRACT NAME SHOULD USE PASCALCASE	1	Automated	*Pending Fix*
I005	Informational	MISSING @AUTHOR IN NATSPEC COMMENTS FOR CONTRACT DECLARATION	1	Automated	*Pending Fix*
I006	Informational	MISSING @DEV IN NATSPEC COMMENTS FOR CONTRACT DECLARATION	1	Automated	*Pending Fix*
I007	Informational	MISSING @DEV IN NATSPEC COMMENTS FOR FUNCTIONS	5	Automated	*Pending Fix*

S. No.	Severity	Bug Type	Instances	Detection Method	Status
I008	Informational	MISSING @INHERITDOC ON OVERRIDE FUNCTIONS	3	Automated	*Pending Fix*
I009	Informational	MISSING NATSPEC COMMENTS IN SCOPE BLOCKS	16	Automated	*Pending Fix*
I010	Informational	MISSING NATSPEC DESCRIPTIONS FOR PUBLIC VARIABLE DECLARATIONS	2	Automated	*Pending Fix*
I011	Informational	MISSING @NOTICE IN NATSPEC COMMENTS FOR CONSTRUCTORS	1	Automated	*Pending Fix*
I012	Informational	MISSING @NOTICE IN NATSPEC COMMENTS FOR FUNCTIONS	3	Automated	*Pending Fix*
I013	Informational	MISSING PAYABLE IN CALL FUNCTION	1	Automated	*Pending Fix*
I014	Informational	MISSING UNDERSCORE IN NAMING VARIABLES	8	Automated	*Pending Fix*
I015	Informational	NAME MAPPING PARAMETERS	2	Automated	*Pending Fix*
I016	Informational	REDUNDANT STATEMENTS	4	Automated	*Pending Fix*
I017	Informational	REVERT STATEMENTS WITHIN EXTERNAL AND PUBLIC FUNCTIONS CAN BE USED TO PERFORM DOS ATTACKS	17	Automated	*Pending Fix*
I018	Informational	USE SCIENTIFIC NOTATION	1	Automated	*Pending Fix*
G001	Gas	CACHE ADDRESS(THIS) WHEN USED MORE THAN ONCE	2	Automated	*Pending Fix*
G002	Gas	CHEAPER INEQUALITIES IN IF()	5	Automated	*Pending Fix*
G003	Gas	CUSTOM ERRORS TO SAVE GAS	2	Automated	*Pending Fix*
G004	Gas	DEFINE CONSTRUCTOR AS PAYABLE	1	Automated	*Pending Fix*
G005	Gas	EMIT USED IN LOOP	4	Automated	*Pending Fix*
G006	Gas	REVERTING FUNCTIONS CAN BE PAYABLE	7	Automated	*Pending Fix*

S. No.	Severity	Bug Type	Instances	Detection Method	Status
G007	Gas	LONG REQUIRE/REVERT STRINGS	2	Automated	*Pending Fix*
G008	Gas	OPTIMIZING ADDRESS ID MAPPING	2	Automated	*Pending Fix*
G009	Gas	STORAGE VARIABLE CACHING IN MEMORY	4	Automated	*Pending Fix*
G010	Gas	USE SELFBALANCE() INSTEAD OF ADDRESS(THIS).BALANCE	1	Automated	*Pending Fix*

04. Vulnerability Details

Issue Type

UNCHECKED ARRAY LENGTH

S. No.	Severity	Detection Method	Instances
M001	Medium	Automated	4

Description

Ethereum is a very resource-constrained environment. Prices per computational step are orders of magnitude higher than with centralized providers. Moreover, Ethereum miners impose a limit on the total number of Gas consumed in a block. If `array.length` is large enough, the function exceeds the block gas limit, and transactions calling it will never be confirmed.

```
for (uint256 i = 0; i < array.length ; i++) { cosltyFunc(); }
```

This becomes a security issue if an external actor influences `array.length`.
E.g., if an array enumerates all registered addresses, an adversary can register many addresses, causing the problem described above.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_26	XOX Token.sol	L167 - L167	*Pending Fix*
SSP_125010_26	XOX Token.sol	L187 - L187	*Pending Fix*
SSP_125010_26	XOX Token.sol	L234 - L234	*Pending Fix*
SSP_125010_26	XOX Token.sol	L254 - L254	*Pending Fix*

Issue Type

BALANCE EQUALITY

S. No.	Severity	Detection Method	Instances
L001	Low	Automated	1

Description

A common defensive programming technique that is useful in enforcing correct state transitions or validating operations is invariant-checking.
In particular, there is one apparent invariant that may be tempting to use but can in fact be manipulated by external users (regardless of the rules put in place in the smart contract). This is the current ether stored in the contract. (`this.balance`)
There are two ways in which malicious actors can (forcibly) send Ether to a contract without using a payable function or executing any code on the contract therefore compromising the contract logic depending on `this.balance` .
These are `selfdestruct/suicide` and pre-loading the contract with Ether.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_7	XOX Token.sol	L351 - L351	*Pending Fix*

Issue Type

LACK OF ZERO VALUE CHECK IN TOKEN TRANSFERS

S. No.	Severity	Detection Method	Instances
L002	Low	Automated	1

Description

The contract is found to lack a check for zero value transfers before executing the function. This oversight can lead t o failures when interacting with certain ERC20 tokens, such as LEND, which revert transactions when a zero transfer is attempted. This can disrupt the flow of the contract and lead to unexpected behavior.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_22	XOX Token.sol	L337 - L337	*Pending Fix*

Issue Type

MISSING EVENTS

S. No.	Severity	Detection Method	Instances
L003	Low	Automated	5

Description

Events are inheritable members of contracts. When you call them, they cause the arguments to be stored in the trans action's log—a special data structure in the blockchain.
These logs are associated with the address of the contract which can then be used by developers and auditors to ke ep track of the transactions.
The contract was found to be missing these events on the function which would make it difficult or impossible to trac k these transactions off-chain.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_73	XOX Token.sol	L306 - L308	*Pending Fix*
SSP_125010_74	XOX Token.sol	L313 - L315	*Pending Fix*
SSP_125010_75	XOX Token.sol	L367 - L373	*Pending Fix*
SSP_125010_76	XOX Token.sol	L382 - L384	*Pending Fix*
SSP_125010_77	XOX Token.sol	L389 - L391	*Pending Fix*

Issue Type

MISSING ZERO ADDRESS VALIDATION

S. No.	Severity	Detection Method	Instances
L004	Low	Automated	2

Description

The contract is found to lack proper validation for zero address inputs in critical functions, particularly where an 'address' input is expected, like during initialization or setting ownership. Failure to validate zero addresses can result in unwanted contract states, such as having critical roles or operations reference the zero address (0×0), which is universally considered an invalid and non-operational address within Ethereum, potentially leading to the loss of assets or control over the contract's functionality.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_19	XOX Token.sol	L153 - L198	*Pending Fix*
SSP_125010_20	XOX Token.sol	L220 - L282	*Pending Fix*

Issue Type

NONREENTRANT MODIFIER PLACEMENT

S. No.	Severity	Detection Method	Instances
L005	Low	Automated	2

Description

When using the `nonReentrant` modifier, it is crucial to place it before all other modifiers in a function. Placing it first ensures that all other modifiers are unable to bypass the reentrancy protection.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_91	XOX Token.sol	L329 - L329	*Pending Fix*
SSP_125010_92	XOX Token.sol	L349 - L349	*Pending Fix*

Issue Type

OUTDATED COMPILER VERSION

S. No.	Severity	Detection Method	Instances
L006	Low	Automated	1

Description

Using an outdated compiler version can be problematic especially if there are publicly disclosed bugs and issues that affect the current compiler version.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_97	XOX Token.sol	L2 - L2	*Pending Fix*

Issue Type

ADDING A RETURN STATEMENT WHEN THE FUNCTION DEFINES A NAMED RETURN VARIABLE IS REDUNDANT

S. No.	Severity	Detection Method	Instances
I001	Informational	Automated	1

Description

The contract defines a named return despite having a return statement. This causes redundancy and can lead to pot ential confusion in the code.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_13	XOX Token.sol	L293 - L297	*Pending Fix*

Issue Type

BLOCK VALUES AS A PROXY FOR TIME

S. No.	Severity	Detection Method	Instances
I002	Informational	Automated	4

Description

Contracts often need access to time values to perform certain types of functionality. Values such as `block.timestamp` and `block.number` can be used to determine the current time or the time delta. However, they are not recommended for most use cases.

For `block.number`, as Ethereum block times are generally around 14 seconds, the delta between blocks can be predicted. The block times, on the other hand, do not remain constant and are subject to change for a number of reasons, e.g., fork reorganizations and the difficulty bomb.

Due to variable block times, `block.number` should not be relied on for precise calculations of time.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_46	XOX Token.sol	L162 - L162	*Pending Fix*
SSP_125010_46	XOX Token.sol	L229 - L229	*Pending Fix*
SSP_125010_47	XOX Token.sol	L295 - L295	*Pending Fix*
SSP_125010_48	XOX Token.sol	L296 - L296	*Pending Fix*

Issue Type

CONSIDER USING UINT48 FOR TIME-RELATED VARIABLES

S. No.	Severity	Detection Method	Instances
I003	Informational	Automated	1

Description

The contract defines time-related variables as `uint256`. Since Unix timestamps can be adequately represented usin g `uint48`, utilizing `uint256` for these variables results in unnecessary gas consumption due to larger storage require ments.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_15	XOX Token.sol	L294 - L294	*Pending Fix*

Issue Type

CONTRACT NAME SHOULD USE PASCALCASE

S. No.	Severity	Detection Method	Instances
I004	Informational	Automated	1

Description

In Solidity, it's advisable to follow the established style guide, which dictates that contract names should be in PascalCase and match their respective file names. This adherence to consistent naming conventions not only enhances code readability and maintainability but also facilitates smoother collaboration among developers. By aligning with these recommendations, you make your code more accessible to others and reduce the likelihood of naming conflicts, contributing to a well-organized and efficient coding environment.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_12	XOX Token.sol	L29 - L29	⚠️ *Pending Fix*

Issue Type

MISSING @AUTHOR IN NATSPEC COMMENTS FOR CONTRACT DECLARATION

S. No.	Severity	Detection Method	Instances
I005	Informational	Automated	1

Description

Natspec `@author` tags are missing from contract declarations in the code. This reduces code clarity and makes it difficult to determine the original author or organization responsible for the contract. The absence of an `@author` tag can lead to challenges in tracking ownership, verifying authenticity, and maintaining proper documentation.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_67	XOX Token.sol	L29 - L392	⚠️ *Pending Fix*

Issue Type

MISSING @DEV IN NATSPEC COMMENTS FOR CONTRACT DECLARATION

S. No.	Severity	Detection Method	Instances
I006	Informational	Automated	1

Description

Natspec `@dev` tags are missing from contract declarations in the code. This reduces code clarity and makes it difficult for developers to understand the contract's internal logic, implementation details, and potential caveats.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_34	XOX Token.sol	L29 - L392	*Pending Fix*

Issue Type

MISSING @DEV IN NATSPEC COMMENTS FOR FUNCTIONS

S. No.	Severity	Detection Method	Instances
I007	Informational	Automated	5

Description

Natspec `@dev` comments are missing from a function declaration in the code. The `@dev` tag provides a detailed explanation of a function's purpose, assumptions, and behavior. Lack of `@dev` documentation can: reduce code maintainability, making it harder for developers to understand the function's implementation details.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_68	XOX Token.sol	L129 - L135	*Pending Fix*
SSP_125010_69	XOX Token.sol	L293 - L297	*Pending Fix*
SSP_125010_70	XOX Token.sol	L306 - L308	*Pending Fix*
SSP_125010_71	XOX Token.sol	L313 - L315	*Pending Fix*
SSP_125010_72	XOX Token.sol	L329 - L340	*Pending Fix*

Issue Type

MISSING @INHERITDOC ON OVERRIDE FUNCTIONS

S. No.	Severity	Detection Method	Instances
I008	Informational	Automated	3

Description

Overridden functions in Solidity should include the `@inheritdoc` tag in their NatSpec comments to ensure documentation consistency and clarity. Failing to include `@inheritdoc` can lead to incomplete or missing documentation, making it difficult for developers and auditors to understand the function's intended behavior. This lack of clarity can increase maintenance overhead and potentially lead to misinterpretations of contract logic.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_16	XOX Token.sol	L107 - L113	*Pending Fix*
SSP_125010_17	XOX Token.sol	L382 - L384	*Pending Fix*
SSP_125010_18	XOX Token.sol	L389 - L391	*Pending Fix*

Issue Type

MISSING NATSPEC COMMENTS IN SCOPE BLOCKS

S. No.	Severity	Detection Method	Instances
I009	Informational	Automated	16

Description

NatSpec comments are missing from an unnamed scope block. Scope blocks can group computations or temporary l ogic and should be documented to clarify purpose and expected behavior. Lack of NatSpec documentation can redu ce readability, making it hard to understand why the scope block exists.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_78	XOX Token.sol	L167 - L181	*Pending Fix*
SSP_125010_79	XOX Token.sol	L180 - L180	*Pending Fix*
SSP_125010_79	XOX Token.sol	L196 - L196	*Pending Fix*
SSP_125010_79	XOX Token.sol	L247 - L247	*Pending Fix*
SSP_125010_79	XOX Token.sol	L280 - L280	*Pending Fix*
SSP_125010_80	XOX Token.sol	L187 - L197	*Pending Fix*
SSP_125010_81	XOX Token.sol	L234 - L248	*Pending Fix*
SSP_125010_82	XOX Token.sol	L238 - L245	*Pending Fix*
SSP_125010_83	XOX Token.sol	L242 - L244	*Pending Fix*
SSP_125010_84	XOX Token.sol	L254 - L281	*Pending Fix*
SSP_125010_85	XOX Token.sol	L259 - L262	*Pending Fix*
SSP_125010_86	XOX Token.sol	L260 - L260	*Pending Fix*
SSP_125010_86	XOX Token.sol	L266 - L266	*Pending Fix*

Bug ID	File Location	Line No.	Action Taken
SSP_125010_86	XOX Token.sol	L271 - L271	⚠️ *Pending Fix*
SSP_125010_87	XOX Token.sol	L264 - L268	⚠️ *Pending Fix*
SSP_125010_88	XOX Token.sol	L269 - L273	⚠️ *Pending Fix*

Issue Type

MISSING NATSPEC DESCRIPTIONS FOR PUBLIC VARIABLE DECLARATIONS

S. No.	Severity	Detection Method	Instances
I010	Informational	Automated	2

Description

Public variable declarations without NatSpec descriptions are found in the code. NatSpec comments improve readability and provide clear documentation for externally accessible variables. Without them, developers and users may struggle to understand the role and impact of these variables when interacting with the contract.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_93	XOX Token.sol	L97 - L97	*Pending Fix*
SSP_125010_94	XOX Token.sol	L98 - L98	*Pending Fix*

Issue Type

MISSING @NOTICE IN NATSPEC COMMENTS FOR CONSTRUCTORS

S. No.	Severity	Detection Method	Instances
I011	Informational	Automated	1

Description

Natspec `@notice` comments are missing from a constructor declaration in the code. The `@notice` tag provides an o verview of what the constructor does, helping users and developers quickly understand its purpose.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_21	XOX Token.sol	L107 - L113	*Pending Fix*

Issue Type

MISSING @NOTICE IN NATSPEC COMMENTS FOR FUNCTIONS

S. No.	Severity	Detection Method	Instances
I012	Informational	Automated	3

Description

Natspec `@notice` comments are missing from a function declaration in the code. The `@notice` tag provides an over view of what the function does, helping users and developers quickly understand its purpose.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_99	XOX Token.sol	L367 - L373	*Pending Fix*
SSP_125010_100	XOX Token.sol	L382 - L384	*Pending Fix*
SSP_125010_101	XOX Token.sol	L389 - L391	*Pending Fix*

Issue Type

MISSING PAYABLE IN CALL FUNCTION

S. No.	Severity	Detection Method	Instances
I013	Informational	Automated	1

Description

The contract is using a `.call()` method to make external calls along with passing some Ether as `msg.value`. Since the function is not marked as `payable`, the transaction might fail if the contract does not have ETH.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_45	XOX Token.sol	L354 - L354	⚠️ *Pending Fix*

Issue Type

MISSING UNDERSCORE IN NAMING VARIABLES

S. No.	Severity	Detection Method	Instances
I014	Informational	Automated	8

Description

Solidity style guide suggests using underscores as the prefix for non-external functions and state variables (private o r internal) but the contract was not found to be following the same.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_37	XOX Token.sol	L37 - L37	*Pending Fix*
SSP_125010_38	XOX Token.sol	L40 - L40	*Pending Fix*
SSP_125010_39	XOX Token.sol	L43 - L43	*Pending Fix*
SSP_125010_40	XOX Token.sol	L68 - L68	*Pending Fix*
SSP_125010_41	XOX Token.sol	L75 - L75	*Pending Fix*
SSP_125010_42	XOX Token.sol	L76 - L76	*Pending Fix*
SSP_125010_43	XOX Token.sol	L97 - L97	*Pending Fix*
SSP_125010_44	XOX Token.sol	L98 - L98	*Pending Fix*

Issue Type

NAME MAPPING PARAMETERS

S. No.	Severity	Detection Method	Instances
I015	Informational	Automated	2

Description

After Solidity 0.8.18, a feature was introduced to name mapping parameters. This helps in defining a purpose for each mapping and makes the code more descriptive.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_1	XOX Token.sol	L68 - L68	*Pending Fix*
SSP_125010_2	XOX Token.sol	L76 - L76	*Pending Fix*

Issue Type

REDUNDANT STATEMENTS

S. No.	Severity	Detection Method	Instances
I016	Informational	Automated	4

Description

The contract contains redundant statements where types or identifiers are declared but not used, leading to no action being performed with them. These statements do not generate any code and can be safely removed to clean up the contract.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_3	XOX Token.sol	L167 - L181	*Pending Fix*
SSP_125010_4	XOX Token.sol	L187 - L197	*Pending Fix*
SSP_125010_5	XOX Token.sol	L234 - L248	*Pending Fix*
SSP_125010_6	XOX Token.sol	L254 - L281	*Pending Fix*

Issue Type

REVERT STATEMENTS WITHIN EXTERNAL AND PUBLIC FUNCTIONS CAN BE USED TO PERFORM DOS ATTACKS

S. No.	Severity	Detection Method	Instances
I017	Informational	Automated	17

Description

In public and external functions, improper use of `revert` can be exploited for Denial of Service (DoS) attacks. An attacker can intentionally trigger these 'revert' conditions, causing legitimate transactions to consistently fail. For example, if a function relies on specific conditions from user input or contract state, an attacker could manipulate these to continually force reverts, blocking the function's execution.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_49	XOX Token.sol	L130 - L130	*Pending Fix*
SSP_125010_50	XOX Token.sol	L131 - L131	*Pending Fix*
SSP_125010_51	XOX Token.sol	L158 - L158	*Pending Fix*
SSP_125010_51	XOX Token.sol	L225 - L225	*Pending Fix*
SSP_125010_52	XOX Token.sol	L159 - L159	*Pending Fix*
SSP_125010_52	XOX Token.sol	L226 - L226	*Pending Fix*
SSP_125010_53	XOX Token.sol	L171 - L171	*Pending Fix*
SSP_125010_54	XOX Token.sol	L172 - L172	*Pending Fix*

Bug ID	File Location	Line No.	Action Taken
SSP_125010_55	XOX Token.sol	L174 - L174	⚠️ *Pending Fix*
SSP_125010_56	XOX Token.sol	L177 - L177	⚠️ *Pending Fix*
SSP_125010_57	XOX Token.sol	L183 - L183	⚠️ *Pending Fix*
SSP_125010_58	XOX Token.sol	L239 - L239	⚠️ *Pending Fix*
SSP_125010_59	XOX Token.sol	L250 - L250	⚠️ *Pending Fix*
SSP_125010_60	XOX Token.sol	L330 - L330	⚠️ *Pending Fix*
SSP_125010_61	XOX Token.sol	L334 - L334	⚠️ *Pending Fix*
SSP_125010_62	XOX Token.sol	L351 - L351	⚠️ *Pending Fix*
SSP_125010_63	XOX Token.sol	L355 - L355	⚠️ *Pending Fix*

Issue Type

USE SCIENTIFIC NOTATION

S. No.	Severity	Detection Method	Instances
I018	Informational	Automated	1

Description

Although the Solidity compiler can optimize exponentiation, it is recommended to prioritize idioms not reliant on com piler optimization. Utilizing scientific notation enhances code clarity, making it more self-explanatory and aligning wit h best practices in Solidity development.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_98	XOX Token.sol	L37 - L37	*Pending Fix*

Issue Type

CACHE ADDRESS(THIS) WHEN USED MORE THAN ONCE

S. No.	Severity	Detection Method	Instances
G001	Gas	Automated	2

Description

The repeated usage of `address(this)` within the contract could result in increased gas costs due to multiple executions of the same computation, potentially impacting efficiency and overall transaction expenses.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_65	XOX Token.sol	L333 - L333	*Pending Fix*
SSP_125010_66	XOX Token.sol	L350 - L350	*Pending Fix*

Issue Type

CHEAPER INEQUALITIES IN IF()

S. No.	Severity	Detection Method	Instances
G002	Gas	Automated	5

Description

The contract was found to be doing comparisons using inequalities inside the if statement.
When inside the `if` statements, non-strict inequalities `(>=, <=)` are usually cheaper than the strict equalities `(>, <)`.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_8	XOX Token.sol	L131 - L131	*Pending Fix*
SSP_125010_9	XOX Token.sol	L159 - L159	*Pending Fix*
SSP_125010_9	XOX Token.sol	L226 - L226	*Pending Fix*
SSP_125010_10	XOX Token.sol	L183 - L183	*Pending Fix*
SSP_125010_11	XOX Token.sol	L250 - L250	*Pending Fix*

Issue Type

CUSTOM ERRORS TO SAVE GAS

S. No.	Severity	Detection Method	Instances
G003	Gas	Automated	2

Description

The contract was found to be using `revert()` statements. Since Solidity `v0.8.4`, custom errors have been introduced which are a better alternative to the revert.
This allows the developers to pass custom errors with dynamic data while reverting the transaction and also making the whole implementation a bit cheaper than using `revert`.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_96	XOX Token.sol	L383 - L383	⚠️ *Pending Fix*
SSP_125010_96	XOX Token.sol	L390 - L390	⚠️ *Pending Fix*

Issue Type

DEFINE CONSTRUCTOR AS PAYABLE

S. No.	Severity	Detection Method	Instances
G004	Gas	Automated	1

Description

Developers can save around 10 opcodes and some gas if the constructors are defined as payable.
However, it should be noted that it comes with risks because payable constructors can accept ETH during deployment.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_14	XOX Token.sol	L107 - L113	*Pending Fix*

Issue Type

EMIT USED IN LOOP

S. No.	Severity	Detection Method	Instances
G005	Gas	Automated	4

Description

In Solidity, when a code emits an event inside of a loop, internally it performs a LOG operation N times, where N repre sents the number of iterations in the loop. This can lead to inflated gas costs and potentially impact the efficiency of the code.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_23	XOX Token.sol	L194 - L194	*Pending Fix*
SSP_125010_23	XOX Token.sol	L278 - L278	*Pending Fix*
SSP_125010_24	XOX Token.sol	L264 - L268	*Pending Fix*
SSP_125010_25	XOX Token.sol	L269 - L273	*Pending Fix*

Issue Type

REVERTING FUNCTIONS CAN BE PAYABLE

S. No.	Severity	Detection Method	Instances
G006	Gas	Automated	7

Description

If a function modifier such as `onlyOwner` is used, the function will revert if a normal user tries to pay the function. Marking the function as payable will lower the gas cost for legitimate callers because the compiler will not include checks for whether a payment was provided.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_27	XOX Token.sol	L129 - L135	*Pending Fix*
SSP_125010_28	XOX Token.sol	L153 - L198	*Pending Fix*
SSP_125010_29	XOX Token.sol	L220 - L282	*Pending Fix*
SSP_125010_30	XOX Token.sol	L306 - L308	*Pending Fix*
SSP_125010_31	XOX Token.sol	L313 - L315	*Pending Fix*
SSP_125010_32	XOX Token.sol	L329 - L340	*Pending Fix*
SSP_125010_33	XOX Token.sol	L349 - L358	*Pending Fix*

Issue Type

LONG REQUIRE/REVERT STRINGS

S. No.	Severity	Detection Method	Instances
G007	Gas	Automated	2

Description

The require() and revert() functions take an input string to show errors if the validation fails.
This strings inside these functions that are longer than 32 bytes require at least one additional MSTORE, along with additional overhead for computing memory offset, and other parameters.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_64	XOX Token.sol	L383 - L383	*Pending Fix*
SSP_125010_64	XOX Token.sol	L390 - L390	*Pending Fix*

Issue Type

OPTIMIZING ADDRESS ID MAPPING

S. No.	Severity	Detection Method	Instances
G008	Gas	Automated	2

Description

Combining multiple address/ID mappings into a single mapping using a struct enhances storage efficiency, simplifies code, and reduces gas costs, resulting in a more streamlined and cost-effective smart contract design.
It saves storage slot for the mapping and depending on the circumstances and sizes of types, it can avoid a Gsset (2 0000 gas) per mapping combined. Reads and subsequent writes can also be cheaper when a function requires both values and they fit in the same storage slot.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_35	XOX Token.sol	L68 - L68	*Pending Fix*
SSP_125010_36	XOX Token.sol	L76 - L76	*Pending Fix*

Issue Type

STORAGE VARIABLE CACHING IN MEMORY

S. No.	Severity	Detection Method	Instances
G009	Gas	Automated	4

Description

The contract is using the state variable multiple times in the function.
SLOADs are expensive (100 gas after the 1st one) compared to MLOAD / MSTORE (3 gas each).

Bug ID	File Location	Line No.	Action Taken
SSP_125010_89	XOX Token.sol	L153 - L198	*Pending Fix*
SSP_125010_89	XOX Token.sol	L153 - L198	*Pending Fix*
SSP_125010_90	XOX Token.sol	L220 - L282	*Pending Fix*
SSP_125010_90	XOX Token.sol	L220 - L282	*Pending Fix*

Issue Type

USE SELFBALANCE() INSTEAD OF ADDRESS(THIS).BALANCE

S. No.	Severity	Detection Method	Instances
G010	Gas	Automated	1

Description

In Solidity, efficient use of gas is paramount to ensure cost-effective execution on the Ethereum blockchain. Gas can be optimized when obtaining contract balance by using `selfbalance()` rather than `address(this).balance` because it bypasses gas costs and refunds, which are not required for obtaining the contract's balance.

Bug ID	File Location	Line No.	Action Taken
SSP_125010_95	XOX Token.sol	L350 - L350	*Pending Fix*

05. Scan History

● Critical ● High ● Medium ● Low ● Informational ● Gas

No	Date	Security Score	Scan Overview
1.	2026-02-24	**91.70**	● 0 ● 0 ● 1 ● 12 ● 64 ● 24

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